

AI Literacy for Mathematical Reasoning

Upon successful completion of the learning tasks, you should be able to:

- LO1: Describe how large language models generate mathematical text.
- LO2: Detect hallucinations in chatbot proofs using the autopsy procedure (labels 0-6).
- LO3: Improve weak prompts using the four parts of a strong prompt.
- LO4: Collaborate with a chatbot using the collaboration procedure (labels 1-6), and evaluate the system behind it.

Resources: Exercise packet: Working with AI on Proofs (Exercises A-C); the course chatbot

Reminders

- Lab 3 Part 2 due Sunday, August 2nd
- Homework 6 due Sunday, August 2nd
- Final Exam Monday, August 3rd at 3:30pm-6pm

Review session on Friday from 1-2pm

Two Patterns of Chatbot Use

Prior work identifies two patterns of student chatbot use on proof and counting problems. The patterns are associated with different assignment outcomes.

Pattern	Description	Average assignment outcome
Cognitive scaffolding	Students interrogate steps, verify small cases, and write the final proof themselves.	Higher Grade%
<u>Dependence</u>	Students paste the problem and copy the answer.	<u>Lower Grade%</u>

This lecture presents three procedures, each decomposed into numbered subgoal labels. The same labels appear in Exercises A–C and apply to assignment work.

Module 1: Next-Token Prediction

A large language model generates text one token at a time. Each token is drawn from a probability distribution conditioned on the preceding text.

Prompt: “Prove that for all $n \geq 1$, ...”

Next-token candidate	Probability
the	<u>0.42</u>
we	0.31
by	0.18
sum	0.05

→

} = 1

No step of this process checks logical validity. The model selects tokens that are statistically plausible given the surrounding text. Training data contains many correct proofs, so plausible output usually resembles a correct proof. When a problem deviates from standard formulations, the statistically plausible continuation and the logically valid continuation differ. A confident, textbook tone is a property of the training data, not evidence of correctness.

$$\underline{T_0} = (\underline{\text{Prove, that, for, all, } n, \geq, 1})$$

$$\underline{T_{i+1}} = (\underline{(T_i), n})$$

$$P(\underline{n = the} | \underline{T_0}) = \underline{0.42}$$

← Choose highest probability

$$P(\underline{n = we} | \underline{T_0}) = \underline{0.31}$$

$$P(n = by | T_0) = 0.18$$

$$P(n = sum | T_0) = 0.05$$

$$\underline{T_1} = (\underline{\text{Prove, that, for, all, } n, \geq, 1, \text{ the}})$$

$$P(\underline{n = integer} | T_1) = \underline{0.95}$$

$$T_2 = (\text{Prove that for all } n \geq 1, \text{ the integer})$$

⋮

$$\underline{T_j} = \text{finished proof}$$

Your prompt:

$T_0' = (\text{Prove that for all integers } n, \text{ if } n, \text{ seven, } n^2 \text{ is even}).$

T_s' = Solved proof

$$P(\underline{n=x} \mid \underline{T_0'}) = \frac{P(\underline{n=x} \wedge \underline{T_0'})}{P(\underline{T_0'})}$$

$P(T_0')$ is related to the number of times

T_0' has appeared in the models training data!

$$P(\underline{n=Let} \mid T_0') = \underline{0.6} \quad \text{highest}$$

$$\underline{T_1'} = (\underline{T_0'}, \underline{Let})$$

$T_1' =$ (Prove that for all integers n , if n is even, n^2 is even. Let)

$$P(n = "n" | T_1') = \frac{P(n = "n" \wedge T_1')}{P(T_1')}$$

$$= \frac{P(n = "n" \wedge T_1')}{\left(\frac{P(n = "let" \wedge T_0')}{P(T_0')} \right)}$$

$$= \frac{P(n = "n" \wedge T_1') \cdot P(T_0')}{P(n = "let" \wedge T_0')}$$

Bayesian Modeling

Accuracy of each term effects sequence representing response to Prompt T_0' .

→ need a lot of datapoints for output to be accurate

need $\frac{P(n = "n" \wedge T_1')}{\text{to all be accurate}}$, $\frac{P(n = "let" \wedge T_0')}{\text{to all be accurate}}$, $\frac{P(T_0')}{\text{to all be accurate}}$

Module 1: How the Model Is Trained

1. Pretraining. The model is adjusted to predict the next token over a large corpus of text and code: textbooks, websites, forums, and papers. Its parameters store the statistical regularities of that corpus. No component stores rules of inference. $\Rightarrow$
2. Post-training. The model is further adjusted, using human feedback, to produce helpful, well-formatted, confident responses. This stage shapes tone and format, not mathematical correctness. $\rightarrow$

Consequence. The reliability of an answer tracks the frequency of the problem’s formulation in the training corpus. Frequent formulations are reproduced accurately. Rare formulations are completed by interpolation toward the nearest frequent pattern, which produces hallucinations. A perturbed problem is a standard problem modified so that it moves from a high-frequency to a low-frequency region of the corpus, with difficulty unchanged for a human solver. Which formulations are frequent reflects which textbooks and examples dominate the corpus.

Standard form (high frequency)	Perturbed form (low frequency)
Prove by induction that $1 + 2 + \cdots + n = \frac{n(n + 1)}{2}$	Prove by induction that $3 + 7 + 11 + \cdots + (4n - 1) = n(2n + 1)$
This formulation appears many times in the training corpus.	The same structure with different coefficients, rare in the corpus. Models substitute an incorrect closed form or revert to the Gauss sum.
Typically answered correctly.	Frequently answered incorrectly.

Module 1: Thinking Models

Recent chatbots offer *thinking* (or *reasoning*) modes. Two things change relative to the description above.

1. **A third training stage.** After pretraining and post-training, the model is further adjusted by reinforcement learning: it is rewarded for producing chains of intermediate reasoning that end in answers which can be checked automatically, as in mathematics and programming. This tunes the model toward reasoning patterns that tend to reach verifiable answers.
2. **More computation at response time.** Before emitting an answer, the model generates intermediate reasoning tokens: it states candidate steps, checks them against each other, and sometimes backtracks, all in generated text. Spending more tokens on this stage generally raises accuracy on mathematical problems.

What does not change. Every token of the reasoning trace is produced by the same mechanism as before: next-token prediction conditioned on the preceding text. The trace is generated text, not a record of an internal derivation, and the version displayed to you may be a summary; reading it is therefore not a substitute for verification. Thinking modes lower the failure rate, including on perturbed problems, but the failure modes of Module 2 persist, and a trace can arrive at a correct final answer through an invalid intermediate step. The procedures in this lecture apply unchanged: a visible trace gives you more material for labels A2 and A3, and label A4's independent check, including its agreement clause, remains necessary.

Reinforcement Learning: reward agent for "correct" decisions

if Agent receives proof prompt:

Prove by induction that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

↓

Looses points for outputting answer directly

6 ans points for outputting plan first

- → Solve Base Case →
 - State inductive hypothesis
 - Solve inductive step
 - final statement
-

repeat until reward function induces intended "correct"
Output

Module 2: Three Failure Modes

- 1. **Axiom reversion.** The model substitutes the standard theorem from which the problem was derived. *Indicator:* the closed form matches the textbook version rather than the problem as stated.
- 2. **Invalid syntactic shortcut.** The model manipulates the form of an expression as though a valid algebraic rule applied, producing a step that is well-formed but false. *Indicator:* a step that appears to follow but fails when checked symbolically.
- 3. **Spatial / counting miscount.** Under constraints such as adjacency or board geometry, the count includes configurations that violate the constraint, or counts symmetric cases more than once. *Indicator:* the case breakdown does not sum to the stated total.

Exercise A (As Group) applies the autopsy procedure, defined below, to transcripts exhibiting these failure modes.

Module 2: The Autopsy Procedure (Subgoal Labels A0–A6)

Apply the labels in order to any transcript suspected of containing a hallucination. Autopsy labels carry the prefix A; the collaboration procedure later in this document uses the prefix C.

- A1. Restate the claim as given. The hypothesis to assume at $n = k$ is $2 + 5 + \dots + (3k - 1) = \frac{k(3k+1)}{2}$.
- A2. Extract the rule the chatbot actually used. It substituted $\frac{k(k+1)}{2}$, the Gauss-sum closed form.
- A3. Locate the first divergence. Line 2 of the inductive step: $\frac{k(k+1)}{2}$ appears where $\frac{k(3k+1)}{2}$ belongs.
- A4. Test a small case. $n = 2$: the claim gives $2 + 5 = 7$ and $\frac{2(7)}{2} = 7$, which agree; the chatbot's final form $\frac{n(n+1)}{2}$ gives 3, which does not.
- A5. Classify the failure mode. Axiom reversion.
- A6. Explain why the failure was likely. The Gauss sum is common in training data; the model substituted the familiar closed form matching the visual pattern of the sum.

Module 2: Worked Example: The Autopsy Procedure

Prompt: Prove by induction: $2 + 5 + 8 + \cdots + (3n - 1) = \frac{n(3n + 1)}{2}$ for all $n \geq 1$.

Chatbot (abridged): Base case ($n = 1$): LHS = 2, RHS = $\frac{1 \cdot 4}{2} = 2$. Inductive step: assume true for $n = k$. Then

$$2 + 5 + \cdots + (3k - 1) + (3(k + 1) - 1) = \frac{k(k + 1)}{2} + (3k + 2) = \cdots = \frac{(k + 1)(k + 2)}{2}. \text{ QED}$$

- A1. **Restate the claim as given.** The hypothesis to assume at $n = k$ is $2 + 5 + \cdots + (3k - 1) = \frac{k(3k + 1)}{2}$.
- A2. **Extract the rule the chatbot actually used.** It substituted $\frac{k(k + 1)}{2}$, the Gauss-sum closed form.
- A3. **Locate the first divergence.** Line 2 of the inductive step: $\frac{k(k + 1)}{2}$ appears where $\frac{k(3k + 1)}{2}$ belongs.
- A4. **Test a small case.** $n = 2$: the claim gives $2 + 5 = 7$ and $\frac{2(7)}{2} = 7$, which agree; the chatbot's final form $\frac{n(n + 1)}{2}$ gives 3, which does not.
- A5. **Classify the failure mode.** Axiom reversion.
- A6. **Explain why the failure was likely.** The Gauss sum is common in training data; the model substituted the familiar closed form matching the visual pattern of the sum.

Module 3: Four Ways to Improve a Weak Prompt

A strong prompt has four parts. Each row of the table states the part, a weak version that omits it, and a strong version that includes it.

Part	Weak version	Strong version
B1. <u>Establish context</u>	<i>"Solve this: $2+5+\dots+(3n-1) = ?$"</i>	"In <u>CSC 226, Module 3</u> on induction. <u>I need</u> to prove $2+5+\dots+(3n-1) = \frac{n(3n+1)}{2}$. Show the <u>inductive step in detail</u> ."
B2. <u>Require step-by-step justification</u>	<i><u>"What's the answer?"</u></i>	" <u>For each step, state which rule or prior line it uses, and why it is valid.</u> "
B3. <u>Build in self-questioning</u>	<i>No check requested.</i>	"Before you finish, list <u>three assumptions in your proof</u> and check each against the <u>problem statement</u> ."
B4. <u>Plan the corrective dialogue</u>	<i>"That's wrong."</i>	" <u>Your step at line 3 used $\frac{k(k+1)}{2}$, but my hypothesis is $\frac{k(3k+1)}{2}$. Redo the step with the correct hypothesis.</u> "

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Module 4: The ICAP Modes of Engagement

Four modes of cognitive engagement, ordered by learning gain.

Mode	With a chatbot	Engagement
<u>Passive</u>	Read the chatbot's output and copy it.	Receive information without further activity.
<u>Active</u>	Re-type the <u>chatbot's</u> proof into your own document.	Manipulate the material without adding to it.
<u>Constructive</u>	Rewrite the <u>chatbot's</u> proof in your own words; fix the error you found.	Produce content beyond what was given.
<u>Interactive</u>	Question the chatbot's steps and respond to each claim.	Construct understanding through dialogue.

Learning gain increases down the table. The collaboration procedure requires the constructive and interactive modes.

Module 4: The System Behind the Chatbot

1. **Origin.** The model was built by a company and trained on text and code produced by people who were generally not credited or compensated. *Consider:* whose work is present in an answer, and who benefits from its use.
2. ~~**Training data coverage.**~~ Perturbation succeeds because standard problems are abundant in training data and unusual ones are rare. What counts as standard reflects which textbooks and traditions are represented. *Consider:* whose notation and problems are well represented, and whose are absent.
3. ~~**Tone and authority.**~~ The confident tone of the output is a property of training, not evidence of correctness. *Consider:* on what grounds an answer is trusted, and what evidence would change that judgment.

The Collaboration Procedure (Subgoal Labels C1–C6)

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- C1. Frame the problem and set the rules. Give context, state the problem exactly, forbid the final answer.
 - C2. Elicit one step with its rule named. Ask for the next step and the counting rule it uses.
 - C3. Interrogate the claim. Request a justification or issue a correction; never accept a step silently.
 - C4. Verify against hand-computed cases. Small cases computed by hand decide who is right.
 - C5. Consolidate in your own words. The write-up is yours, not the chatbot's.
 - C6. Question the system. Ask who built it, on whose data, and why you trust its answer.

When to Use the Collaboration Procedure

Use the collaboration procedure when all of the following hold.

1. The problem is unfamiliar to you but within the scope of the course, so working it is a learning task rather than a lookup.
2. You can verify claims independently, for example by computing small cases by hand (label C4 depends on this).
3. The deliverable is understanding expressed in your own write-up, not a copied answer (label C5).
4. Course policy permits AI use on the task.

Do not use the collaboration procedure when course policy prohibits AI use on the task; when you have no independent means of verification, since the procedure's safeguards then have nothing to rest on; or when you only need a definition or a standard formula, since a textbook is faster and does not hallucinate.

The two procedures connect as follows. The collaboration procedure governs the conversation; the autopsy procedure examines a single suspect response. If a collaboration session produces a response you suspect is hallucinated, pause the loop and apply the autopsy procedure (labels A0–A6) to that response, then resume.

Collaboration Example Dialogue

The example below applies the collaboration procedure to the following problem: how many ways can 8 rooks be placed on a standard 8×8 chessboard so that no two attack each other? Each student turn is tagged with the subgoal label it performs.

STUDENT [C1: frame the problem and set the rules]: I'm in CSC 226 working on combinatorics. I want to figure out how many ways 8 non-attacking rooks can be placed on a standard 8×8 chessboard. I have not seen this problem before. Please do NOT give me the final number. Instead, help me set up the counting argument step by step, and for each step tell me what rule of counting you're using and why.

CHATBOT: Two rooks attack each other if they share a row or a column. So for a valid placement, each row must contain exactly one rook, and each column must contain exactly one rook. Step 1 (product rule): choose which column the rook in row 1 goes in — 8 choices. Step 2: for row 2, the column used by row 1 is forbidden, so there are 7 choices. Continuing, row k has $(8 - k + 1)$ choices. The total is a product.

STUDENT [C3: interrogate the claim]: Justify the claim that each row must contain exactly one rook. Why couldn't a valid placement have a row with zero rooks and another row with two?

CHATBOT: A row with two rooks immediately has two rooks sharing that row, so they attack each other. For 8 rooks on 8 rows with no two sharing a row, we need at most one per row; with 8 rooks and 8 rows, each row must contain exactly one. The same argument works for columns.

STUDENT [C2: elicit one step with its rule named]: That argument is clean. Give me the next step only, and name the counting rule it uses.

CHATBOT: Next step (product rule): the per-row choices are independent stages of one construction, so the total is the product $8 \times 7 \times 6 \times \cdots \times 1$.

STUDENT [C3: interrogate the claim]: Now I can see where you're going: the answer is $8!$. Explain to me why this is equivalent to counting permutations of $\{1, \dots, 8\}$. I want to be sure I understand the bijection before I write it up.

The first prompt performs label C1. Each subsequent student turn performs label

C2 or C3. To conclude, the student would verify the argument on a small case such as a 2×2 board (label C4: two placements, matching $2!$), write the bijection argument in their own words (label C5), and note what about the system did or did not earn trust (label C6).

Self-Check: Which subgoal label does the following student turn perform? “Before I accept that, why does the product rule apply here rather than the sum rule?”

Self-Check: A student wants the definition of a permutation before starting homework. Should they use the collaboration procedure?

Summary

1. **A language model predicts text; it does not verify logic.** Each step of chatbot output is a statistical continuation conditioned on the surrounding text, not a derivation.
2. **Verify output against the problem as stated.** Apply the autopsy labels: restate the claim, extract the rule used, test a small case, and check counting arguments against constraints.
3. **Prompt for justification rather than answers.** Establish context, require step-by-step justification, build in self-questioning, and plan the corrective dialogue.
4. **Evaluate the system as well as its output.** Consider who built the chatbot, on what data it was trained, and whose interests its use serves.